

WG03

Natural Vegetation and Biomes

About 1 question a paper. Two things are documented repeats: the temperate grassland names matched to their regions, and the RAS 2024 fact that tropical forests record the maximum deforestation in the world.

Biomes is the friendliest chapter in world geography. It is a set of names - Selva, Llanos, Prairie, Steppe, Veld, Taiga, Tundra - each tied to one region and one kind of plant. Fix the names to the map and the questions answer themselves.

World Biomes

Tropical

- Equatorial Selva
- Monsoon deciduous
- Savanna

Temperate grasslands

- Prairie
- Pampas
- Steppe
- Veld
- Downs
- Canterbury
- Puszta

Mediterranean scrub

- Maquis
- Chaparral
- Matorral
- Fynbos
- Mallee

Cold lands

- Taiga - largest land biome
- Tundra - permafrost

Special habitats

- Desert xerophytes
- Mangrove - Sundarbans

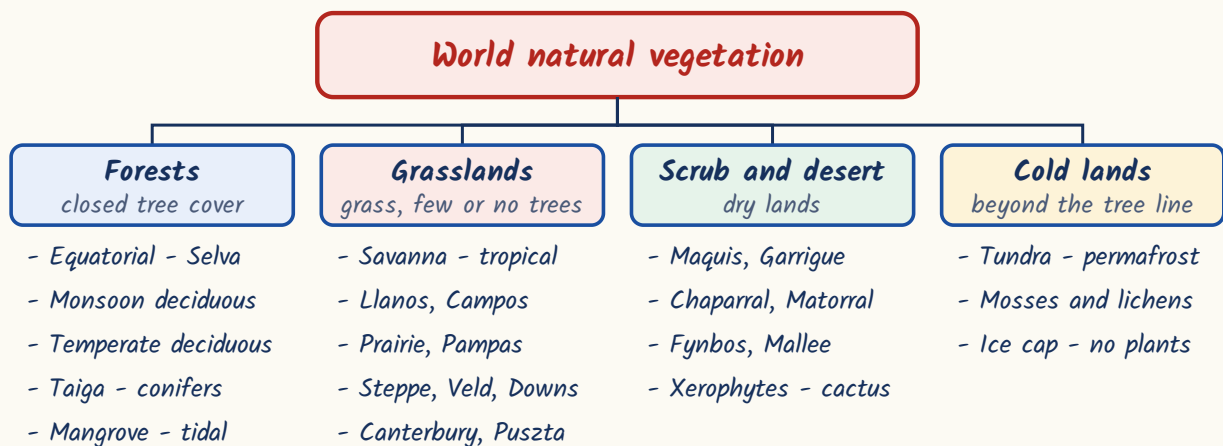
Forest facts

- 4.14 bn ha
- 31-32% of land
- Tropical forests lose most

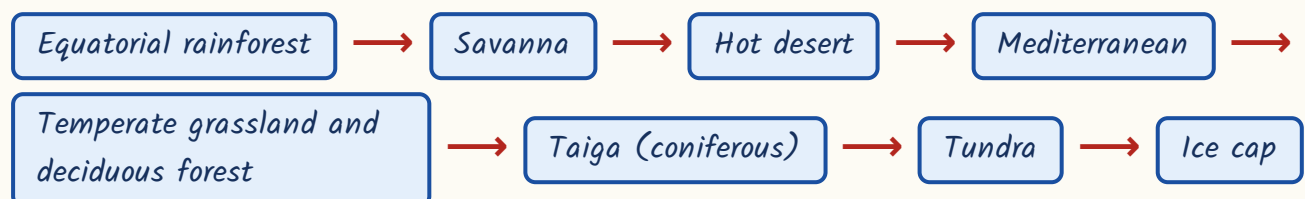
How the biome map is arranged

Vegetation follows climate, and climate follows latitude. So the belts run in a fixed order from the Equator to the pole. Learn that order once and you can place any biome. The same sequence repeats vertically as you climb a mountain - that is why the Himalaya shows tropical forest at the foot and alpine pasture near the top.

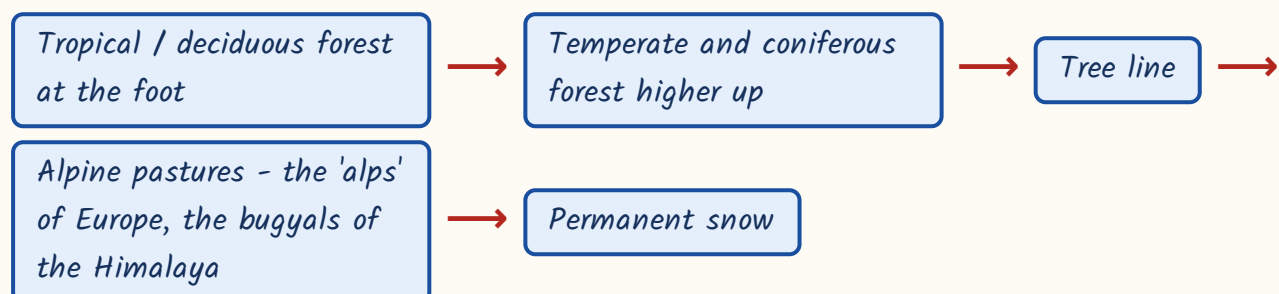
The biomes classified



Vegetation belts from the Equator poleward



The same sequence going up a mountain (altitudinal zonation)



ASKED IN RAS

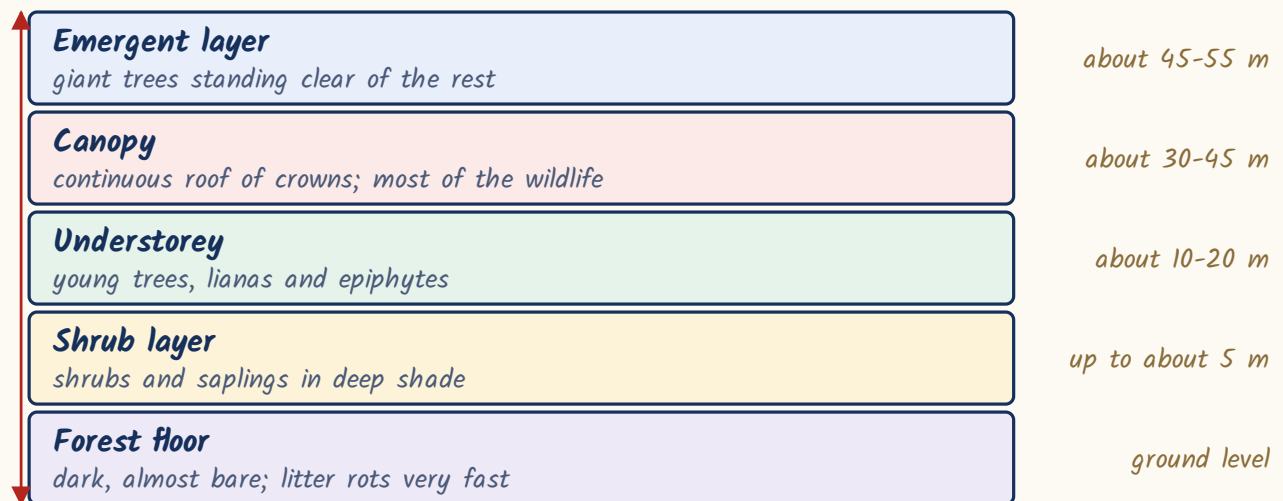
RPSC gives world geography about 3 questions and takes them as matching sets - peaks in 2016, 2018 and 2021, deserts and deforestation in 2024. In this chapter the matching pools are exactly three: temperate grassland names, Mediterranean scrub names, and the tropical trio Selva-Llanos-Campos. Over-learn those three lists.

The tropical belt - rainforest, monsoon forest, savanna

Three biomes share the tropics and they differ by how the rain is spread through the year. Rain all year gives evergreen rainforest. Rain in one season gives deciduous monsoon forest. Still less rain gives grassland with scattered trees - the savanna.

- Equatorial evergreen rainforest is locally called 'Selva' in the Amazon.
- It occurs in the Amazon basin, the Congo basin and South-East Asia.
- Multi-storeyed with a dense canopy, lianas and epiphytes; there is no distinct leaf-shedding season.
- Hardwoods: mahogany, ebony, rosewood, greenheart, cinchona.
- Trees grow scattered in mixed stands, so commercial exploitation is difficult.

Vertical strata of the tropical rainforest



- Tropical deciduous or monsoon forest - teak, sal, sandalwood, bamboo.
- It sheds its leaves in the dry season to save water.
- Commercially it is the most useful forest of the monsoon lands, including India.
- Savanna is tropical grassland - tall coarse grass with scattered acacia and baobab.
- It is called the 'big game country' - lion, zebra, giraffe, elephant, wildebeest.
- Do not confuse the names: Selva is a forest, while Llanos and Campos are grasslands.

Tropical grassland names by region

Name	Region
Llanos	Orinoco basin, Venezuela
Campos	Brazilian highlands
Sudan / Sahel type savanna	Africa
Northern Australian savanna	Australia

Temperate grasslands – the repeat you cannot miss

This is the highest-yield table in the chapter. Every temperate grassland has a local name and one region, and the question is always the pairing. These grasslands lie in continental interiors, get only 25-75 cm of rain, and are practically treeless – which is why they became the world's granaries and its great ranching lands.

Temperate grassland to region

Grassland	Region
Prairie	North America – USA and Canada
Pampas	Argentina and Uruguay
Steppe	Eurasia – Russia, Ukraine, Central Asia
Veld	South Africa
Downs	Australia
Canterbury	New Zealand
Puszta	Hungary
Manchurian	North-east China

- Practically treeless, because rainfall is too low and too seasonal for trees.
- They lie in continental interiors, far from the moderating sea.
- Rainfall 25-75 cm a year.
- They are the world's granaries – wheat and maize – and its great ranching lands.
- The herdsmen of the Pampas are called gauchos.

YAAD RAKHO

Walk the world in order: Prairie (North America) - Pampas (South America) - Steppe (Eurasia) - Veld (Africa) - Downs (Australia) - Canterbury (New Zealand) - Puszta (Hungary). Same first letters as the continents in most cases: P-P-S-V-D-C-P.

ASKED IN RAS

The question bank marks Prairie-North America, Pampas-Argentina, Veld-South Africa and Canterbury-New Zealand as repeat items, and the NOT-correctly-matched version uses 'Pampas - South Africa' as the wrong pair. RAS asks this as a straight match-the-following, so recall must be instant.

TRAP

The two swaps RPSC lives on: Veld is South Africa and Pampas is Argentina - never the other way round. And Downs is the Australian grassland, not the chalk downs of England.

Mediterranean lands and temperate deciduous forest

Mediterranean climate means hot dry summers and mild wet winters - the reverse of the monsoon. Plants survive the dry summer by being sclerophyllous: thick bark, small waxy leaves, deep roots. These are the orchard lands of the world, and each region gives its scrub a different name.

Mediterranean scrub names

Scrub name	Region
Maquis and Garrigue	Mediterranean basin
Chaparral	California

Scrub name	Region
Matorral	Central Chile
Fynbos	Cape region, South Africa
Mallee	South-western Australia

- Typical crops: citrus fruit, olives, grapes and cork oak.
- The region is called the 'orchard land' of the world.
- Adaptations to the dry summer: thick bark, small waxy leaves, deep roots.

- Temperate deciduous forest - oak, ash, beech, elm, maple.
- Found in western Europe, eastern USA and eastern Asia.
- It is the most cleared of all forest belts, because these are the world's old industrial and farming regions.

Cold lands and special habitats

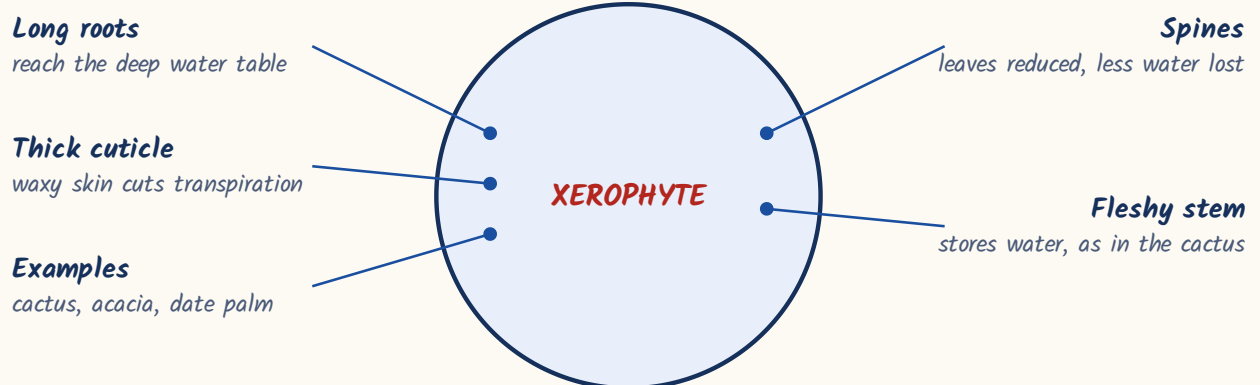
North of the deciduous belt come two biomes that are always asked against each other - taiga and tundra. Taiga is the coniferous forest, tundra is the treeless cold desert beyond the tree line. Both lie almost entirely in the Northern Hemisphere, simply because there is very little land at those latitudes in the south.

Taiga vs Tundra

Taiga (boreal coniferous forest)	Tundra
Largest land biome; a continuous belt across Siberia, Canada and Scandinavia	Lies beyond the tree line, on the Arctic fringe
Conifers - pine, fir, spruce, larch, cedar; evergreen, conical, needle-leaved	Mosses, lichens (reindeer moss) and sedges; no trees
World's chief source of softwood, pulp, newsprint and matchwood	No commercial timber; growing season only about two months
Grizzly bear, moose, lynx, wolf	Caribou or reindeer, musk ox, arctic fox, polar bear
Confined almost wholly to the Northern Hemisphere	Underlain by permafrost - permanently frozen sub-soil

- Desert vegetation is xerophytic - cactus, acacia, date palm, euphorbia.
- Xerophyte adaptations: long roots, thick cuticle, spines in place of leaves, water-storing stems.
- Mangrove or tidal forest is salt tolerant, with stilt roots and pneumatophores for breathing.
- The Sundarbans of the Ganga-Brahmaputra delta is the world's largest mangrove forest.

How a xerophyte survives the desert



- Orangutan - Borneo and Sumatra; gorilla, okapi and bonobo - Congo basin.
- Grizzly bear, moose, lynx and wolf - North American coniferous belt.
- Kangaroo - Australian grassland.
- Coniferous softwood forests are the world's leading lumbering regions.
- Equatorial hardwood forests are lumbered far less - mixed stands and poor accessibility.

YAAD RAKHO

Taiga has the timber, tundra has the turf. Both begin with T; the one with trees is the one whose name has more letters in common with 'timber'.

Forest statistics and deforestation - the 2024 question

This is the one part of the chapter with hard numbers, and RAS 2024 asked it twice in the same paper - once directly and once as a NOT-correct statement. The single fact to burn in: tropical forests record the maximum deforestation in the world. Not temperate, not boreal - tropical.

- World forest area is about 4.14 billion hectares - roughly 31-32 per cent of the global land area.
- More than half the world's forests lie in five countries: Russia, Brazil, Canada, the United States and China.
- Russia alone has about 832 million hectares - about one-fifth of the world total.
- Nearly half of the world's forests are tropical.
- The Amazon, largely in Brazil, holds the world's largest area of tropical rainforest and is called the 'lungs of the earth'.

How the world's forest area is divided



- Global deforestation has slowed - from about 17.6 million hectares a year in the 1990s to about 10.9 million hectares a year in the most recent decade.
- The largest direct driver of tropical deforestation is conversion of forest to agricultural land.
- That means shifting cultivation, cattle ranching in the Amazon, and oil-palm and soy plantations in South-East Asia and Brazil.

ASKED IN RAS

RAS Pre 2024 asked outright: at present the maximum deforestation in the world is taking place in which type of forest? Answer - tropical forests. The same paper's negative version made 'Temperate forests are recording the maximum deforestation in the world' the incorrect statement. One fact, two questions.

CURRENT LINK

Current link: the FAO's Global Forest Resources Assessment 2025, released in October 2025, confirms world forest area at about 4.14 billion hectares and annual forest loss down to about 10.9 million hectares a year for 2015-25, from 17.6 million in the 1990s; South America still loses the most forest each year. As of September 2026 this is the latest FAO assessment - quote its figures, not older ones.

CURRENT LINK

Current link: COP30, the UN climate conference, met at Belem in the Brazilian Amazon in November 2025 - the first COP held inside the Amazon - and made standing tropical forest its headline theme, launching the Tropical Forests Forever Facility, a fund meant to pay tropical countries for forest left standing rather than cut. As of September 2026 Belem is the reference point for any tropical deforestation question.

TRAP

Two number traps. Forests cover about 31 per cent of the LAND area, which is roughly one-third of the land - not one-third of the earth's surface, because most of the earth is ocean. And deforestation is slowing, but it has not stopped; do not read the falling rate as forest gain.

REVISION RECAP

1. Selva = equatorial rainforest of the Amazon; Llanos = Orinoco grassland; Campos = Brazilian highlands grassland.
2. Equatorial hardwoods - mahogany, ebony, rosewood, greenheart, cinchona; scattered stands make logging hard.
3. Monsoon deciduous forest - teak, sal, sandalwood, bamboo; commercially the most useful of the monsoon lands.
4. Savanna = tall coarse grass with acacia and baobab; the 'big game country'.
5. Prairie-North America, Pampas-Argentina, Steppe-Eurasia, Veld-South Africa, Downs-Australia, Canterbury-New Zealand, Puszta-Hungary.
6. Temperate grasslands are treeless, continental, 25-75 cm rain, the world's granaries; Pampas herdsmen are gauchos.
7. Mediterranean scrub: Maquis and Garrigue in the basin, Chaparral in California, Matorral in Chile, Fynbos in the Cape, Mallee in south-west Australia.
8. Mediterranean lands are the world's orchards - citrus, olives, grapes, cork oak; sclerophyllous plants.
9. Taiga is the largest land biome, almost wholly in the Northern Hemisphere, and supplies the world's softwood.

10. Tundra lies beyond the tree line on permafrost - mosses, lichens, sedges; growing season about two months.
11. Temperate deciduous forest (oak, ash, beech, elm, maple) is the most cleared forest belt of all.
12. Sundarbans is the world's largest mangrove; mangroves use stilt roots and pneumatophores.
13. World forests about 4.14 billion ha, 31-32 per cent of land; Russia, Brazil, Canada, USA and China hold over half.
14. TROPICAL forests record the maximum deforestation - the RAS 2024 fact, asked directly and negatively.
15. Deforestation fell from about 17.6 m ha a year in the 1990s to about 10.9 m ha a year now; the driver is farmland conversion.